

CardioTwin India

- Personalized Cardiovascular Risk Digital Twin
- Digital Twin Challenge 2026
- Amity University Noida

The Problem

- Patient risk is dynamic, but clinical records are often snapshots.
- We combine historical clinical context with changing wearable physiology.
- Target outcome: elevated cardiovascular-risk state.

Our Digital Twin

- Static EHR context + dynamic wearable window.
- Twin state is refreshed as new observations arrive.
- Output: risk probability, risk band and explainable drivers.

Two Data Streams

- STATIC: age, BMI, hypertension, diabetes, smoking, SBP, LDL, prior CVD.
- DYNAMIC: heart rate, HRV, sleep, steps, activity and recent baseline change.
- All PoC data is synthetic.

Architecture

- EHR → feature layer
- Wearables → time-series feature layer
- Both → Digital Twin State → model → explanation → dashboard

Machine Learning

- Logistic Regression baseline.
- $P(Y=1|X) = 1 / (1 + e^{-(\beta_0 + \beta^T X)})$
- Transparent, fast and reproducible for a prototype.

Feature Engineering

- Seven-day rolling window.
- Mean HR, resting HR, mean HRV, HRV slope.
- Sleep, steps, activity and heart-rate change.

Dashboard

- Patient selector
- Risk probability and risk band
- Clinical context
- HR/HRV trends
- Feature contributions

Evaluation

- ROC-AUC, accuracy, precision, recall, F1.
- Confusion matrix.
- Synthetic metrics demonstrate pipeline behavior—not clinical efficacy.

Explainability

- Model contributions connect patient features to the risk output.
- Supports clinician inspection rather than black-box alerting.
- Final decisions remain with qualified professionals.

Safety & Privacy

- No real patient-identifiable data.
- Synthetic sandbox for the PoC.
- Future deployment requires validation, security, privacy, bias assessment and regulatory review.

Clinical Workflow

- Wearable observations → Twin update → risk signal → clinician review → clinical decision.
- No autonomous diagnosis or prescribing.

Future Roadmap

- Temporal deep-learning models
- Personal baseline adaptation
- Uncertainty estimation
- External and prospective validation
- EHR interoperability
- Privacy-preserving/federated learning

Demo

- Run: `python run_pipeline.py`
- Run: `streamlit run dashboard/app.py`
- Select virtual patients and inspect how the twin state and risk output change.

Conclusion

- Observe → Update → Predict → Explain → Review
- CardioTwin India demonstrates the core architecture of a localized healthcare Digital Twin PoC.